

Demo Script

Project Demonstration Script — OptiCrop

1. Demo Objective

Showcase OptiCrop's end-to-end flow: from entering soil/environmental data to receiving an ML-generated crop recommendation.

2. Demo Flow

Step 1 — Introduction (1-2 min)

- Briefly state the problem: farmers lack data-driven crop guidance
- Introduce OptiCrop as the solution

Step 2 — Architecture Walkthrough (2 min)

- Show high-level architecture: Frontend → Flask Backend → ML Model → Result
- Mention the 5 algorithms compared (KNN, Logistic Regression, Decision Tree, Random Forest, K-Means) and the selection of the best-performing model

Step 3 — Live Demo (3-5 min)

1. Launch Flask app locally (`python app.py`)
2. Open browser to `http://127.0.0.1:5000/`
3. Enter sample input:
 - N = 90, P = 42, K = 43
 - Temperature = 20.8°C
 - Humidity = 82%
 - pH = 6.5
 - Rainfall = 202 mm
4. Click "Predict"
5. Show the recommended crop on the result page
6. Repeat with a second, contrasting input set to show different recommendations

Step 4 — Edge Case Demo (1-2 min)

- Show input validation in action (e.g., leaving a field blank or entering invalid data)

Step 5 — Model Performance Summary (1-2 min)

- Present accuracy comparison table across the 5 models
- State which model was selected for deployment and why

Step 6 — Closing & Future Scope (1 min)

- Summarize impact: helps farmers make data-driven decisions
- Mention future scope: weather API integration, fertilizer recommendations, cloud deployment

3. Sample Talking Points

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- “OptiCrop takes seven soil and climate parameters and, using a trained machine learning model, recommends the crop most likely to yield the best results under those conditions.”
 - “We compared five different algorithms and selected the one with the highest accuracy for deployment.”

4. Pre-Demo Checklist

- ☐ `model.pkl` present in Flask folder
- ☐ All dependencies installed (`pip install -r requirements.txt`)
- ☐ Flask server runs without errors
- ☐ Sample input values ready and tested beforehand
- ☐ Backup screenshots/recording in case of live failure